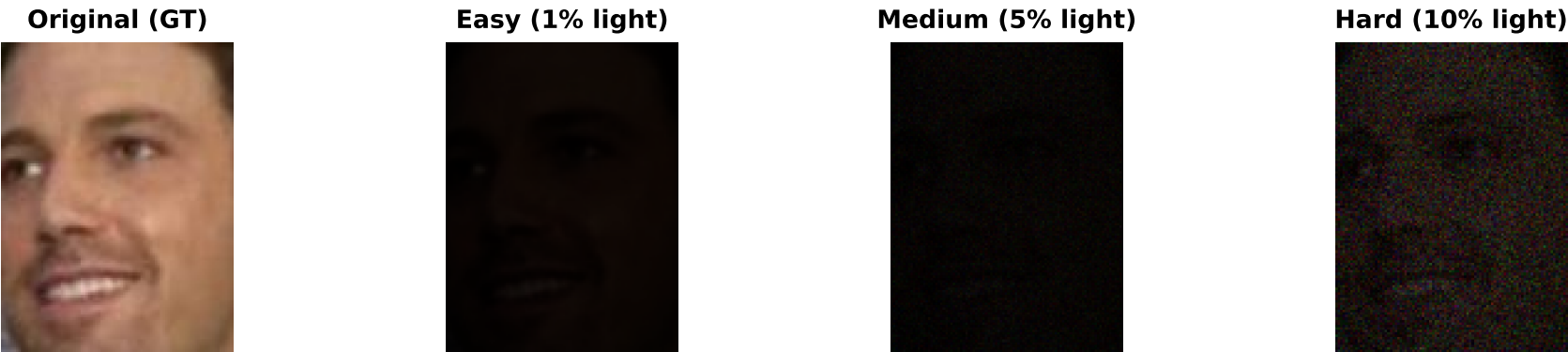


Figure 1: Multi-Level Low-Light Dataset Generation



Physics-Based Synthesis Pipeline:

sRGB Image (Input) $\rightarrow$ Linear RGB (γ^{-1}) $\rightarrow$ Light Reduction ($\times$ reduction_factor) $\rightarrow$ Noise Addition (Poisson-Gaussian) $\rightarrow$ White Balance (Per-channel gain) $\rightarrow$ Linear/sRGB (γ or RAW)

Level	Light %	Noise Type	WB Shift	Raw Sensor Mode	Gamma Correction
Easy	1%	None	No	False	Applied
Medium	5%	Poisson-Gaussian ($\sigma_{\text{shot}}=1.0$, $\sigma_{\text{read}}=0.005$)	No	True	Skipped
Hard	10%	High P-G Noise ($\sigma_{\text{shot}}=2.0$, $\sigma_{\text{read}}=0.015$)	Yes ($\pm 10\%$)	True	Skipped